

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Evaluation of Small Mass Spectrometer Systems as Candidates for the Development of Miniature Mass Spectrometer Systems

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ABSTRACT

There is an ever increasing need for smaller mass spectrometer (MS) systems. Smaller MS systems are generally more easily transported, less expensive, operated in more diverse environments, and allow real-time analysis. Such systems have many potential applications such as environmental analysis, process monitoring, and hazardous chemical detection. At Kennedy Space Center there is a need for small, point-sensor MS systems for a variety of tasks. Such tasks include the monitoring of hazardous gases around the Space Shuttle prior to launch, the detection of toxic vapors, such as hypergols, and process control of a fuel production plant on Mars. Several types of small MS systems are evaluated and discussed, including linear quadrupole, quadrupole ion trap, time of flight and magnetic sector. The performance of each system in terms of accuracy, precision, limits of detection, response time, recovery time, scan rate, volume and weight is assessed. A performance scale is setup for each test category in order to rank the systems and an overall rank is given to each system.